

# DFE-based simulation model for inference of effect size distributions

## Model overview

For each locus  $j$ :

$$s_{ud,j} \sim p_{\text{DFE}}(s_{ud}), \quad X_j \sim p_X(x \mid s_{ud,j}), \quad \beta_{s,j} \mid s_{ud,j} \sim p_\beta(\beta_s \mid s_{ud,j}),$$
$$\text{SE}_j = (2X_j(1 - X_j)n_{\text{eff}})^{-1/2}, \quad \hat{\beta}_{s,j} \mid \beta_{s,j}, \text{SE}_j \sim N(\beta_{s,j}, \text{SE}_j^2).$$

Ascertainment indicator (default MAF mode):

$$\mathbf{1}_j = \mathbf{1}\{\min(X_j, 1 - X_j) \geq m^*\}.$$

Observed tuple exported to `mutvar infer`:

$$(\text{effect\_allele\_frequency}, \text{beta}, \text{standard\_error})_j = (X_j, \hat{\beta}_{s,j}, \text{SE}_j).$$

## Core effect and scaling relationships

Notation:

- $s_{ud,j}$ : underdominant selection coefficient,
- $S_{ud,j} = 2N_e s_{ud,j}$ : scaled underdominant coefficient,
- $X_j \in (0, 1)$ : effect-allele frequency,
- $\beta_{s,j}$ : latent effect on the  $\beta_s$  axis,
- $\hat{\beta}_{s,j}$ : observed effect estimate,
- $\text{SE}_j$ : standard error,
- $n_{\text{eff}}$ : effective GWAS sample size controlling observation noise.

All beta quantities in this document are on the selection-scaled axis  $\beta_s$  (not raw trait-scale effect units). In the one-dimensional limit,  $\beta_s^2 = S_{ud}$ , which fixes the scale of both  $\beta_s$  and  $\hat{\beta}_s$ .

## Practical DFE parameterization for focal-trait effects

In practical runs, a common parameterization expresses focal-trait effects as a mixture of:

1. a DFE-linked zero atom, and
2. a trait-coupling null gate applied on top of the latent draw.

The focal-trait law is written as

$$p_\beta(\beta_s \mid s_{ud}) = p_0 \delta_0(\beta_s) + (1 - p_0) \left[ p_t \delta_0(\beta_s) + (1 - p_t) p_{\beta,+}(\beta_s \mid s_{ud}) \right],$$

with

$$p_0 = \text{dfe.point\_mass\_zero}, \quad p_t = \text{effect.trait\_null\_fraction}.$$

Interpretation:

- $p_0$  controls the DFE-level atom at zero,
- $p_t$  controls an additional focal-trait null fraction among variants not already set to zero by  $p_0$ ,
- $p_{\beta,+}(\cdot \mid s_{ud})$  is the nonzero conditional component parameterized from the SSD table.

Effect model under stabilizing-selection limits:

## High-dimensional (pleiotropic)

$$\beta_{s,j} \mid s_{ud,j} \sim N(0, 2N_e s_{ud,j}) = N(0, S_{ud,j}).$$

### One-dimensional (single-trait)

$$\beta_{s,j} \mid s_{ud,j} \in \left\{ -\sqrt{2N_e s_{ud,j}}, +\sqrt{2N_e s_{ud,j}} \right\} = \left\{ -\sqrt{s_{ud,j}}, +\sqrt{s_{ud,j}} \right\},$$

with equal sign probability.

Observation model:

$$\text{SE}_j = (2X_j(1 - X_j)n_{\text{eff}})^{-1/2}, \quad \hat{\beta}_{s,j} \mid \beta_{s,j}, \text{SE}_j \sim N(\beta_{s,j}, \text{SE}_j^2).$$

### Ascertainment

Default discovery mode is a MAF threshold:

$$\min(X_j, 1 - X_j) \geq m^*.$$

Here  $v_s^*$  denotes the fixed variance-contribution cutoff used by variance-threshold ascertainment.

Optional alternative ascertainment modes use variance contribution thresholds with

$$v_{s,j} = 2X_j(1 - X_j)\beta_{s,j}^2, \quad \hat{v}_{s,j} = 2X_j(1 - X_j)\hat{\beta}_{s,j}^2,$$

and keep loci by either  $v_{s,j} > v_s^*$  (truth-side) or  $\hat{v}_{s,j} > v_s^*$  (noisy-side).

If variance-threshold ascertainment is used with target  $p^*$ , the implied effective GWAS sample size is:

$$n_{\text{eff}} = \frac{2 \operatorname{erf}^{-1}(1 - p^*)^2}{v_s^*}.$$

### Mutvar-facing observed output

Observed output is written for direct input to `mutvar infer` with exactly:

- `effect_allele_frequency`,
- `beta`,
- `standard_error`.

At each locus  $j$ , the observed columns map to latent quantities as:

- `effect_allele_frequency` =  $X_j$ ,
- `beta` =  $\hat{\beta}_{s,j}$ ,
- `standard_error` =  $\text{SE}_j$ .

So the exported `beta` column is on the selection-scaled axis  $\beta_s$ . The exported `standard_error` is on that same  $\beta_s$  axis.

### Technical generation details

The nonzero DFE component is parameterized on

$$\ell = \log_{10}(s_{ud}),$$

using an SSD table  $\{(\ell_i, f_i)\}_{i=1}^m$  with interpolation-defined density

$$p_\ell(\ell) \propto \text{interp}(\ell; \{\ell_i, f_i\}).$$

Allele frequencies are sampled from an underdominant SFS conditional on  $s_{ud}$ . The SFS kernel is parameterized with  $2S_{ud}$ , where  $S_{ud} = 2N_e s_{ud}$ :

$$\log \tau(x \mid S_{ud}) = \log \left( \frac{\theta}{x(1-x)} \right) - (2S_{ud})x(1-x) + \text{erf correction}(x, 2S_{ud}).$$